SOFTWARE REQUIREMENTS SPECIFICATION

CT-365 Software Engineering

Project Title	:	**Bloodbound – Blood Donor & Donation Camp Coordination System**
Course & Code	:	**CT-365 Software Engineering (CCP)**
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Thar Institute of Engineering, Sciences and Technology (TIEST)
A constituent college of NED University of Engineering & Technology

Software Requirements Specification

Bloodbound – Blood Donor & Donation Camp Coordination System

> **Document Purpose:** This specification details the system actors, functional rules, and non-functional engineering standards for the Bloodbound web application under Course CT-365 (Software Engineering).

1. SYSTEM ROLES & ACTORS

- **Donor:** Registers personal details, blood group, location, tracks donation history, calculates health eligibility, and responds to urgent match alerts.
- **Hospital Staff / Patient Requester:** Authorized hospital blood bank personnel or patient relatives/family members creating and monitoring emergency blood requests.
- **Camp Organizer:** Schedules, manages, and tracks community blood donation drives and unit target capacities.
- **System Admin:** Oversees platform security, user moderation, and system data integrity.

2. FUNCTIONAL REQUIREMENTS (FR)

Req ID	Module	Description	Priority
FR-01	User Auth	Users must register and log in securely with role-based routing (Donor, Hospital Staff / Requester, Camp Organizer, Admin).	High
FR-02	Donor Profile	Donors manage personal profiles, blood type, city location, last donation date, and availability status.	High
FR-03	Request Mgmt	Hospital staff / patient requesters submit urgent blood requests specifying blood group, units needed, urgency, and hospital location.	High
FR-04	Matching Engine	System automatically matches emergency requests with eligible donors based on compatible blood groups and city proximity.	High
FR-05	Alert System	Broadcasts notifications and alerts to matched active donors upon emergency request submission.	High
FR-06	Donor Privacy	Donor contact details remain hidden from requesters until the donor explicitly accepts a matching request.	High
FR-07	Camp Mgmt	Camp organizers schedule donation drives specifying venue, operational hours, and target donor capacity.	Medium
FR-08	Slot Booking	Donors pre-book time slots for upcoming donation camps to ensure crowd control.	Medium
FR-09	Eligibility	Automatically calculates donor health eligibility based on the minimum required gap between donations.	High
FR-10	Admin Audit	System administrators oversee account verifications and platform content moderation.	Medium

3. NON-FUNCTIONAL REQUIREMENTS (NFR)

Req ID	Category	Description	Target Metric
NFR-01	Performance	Emergency matching engine and donor alert broadcasts execute rapidly across feeds.	Response time < 2.0s
NFR-02	Security	User passwords must be securely hashed and API endpoints authenticated using session tokens.	bcryptjs + JWT validation
NFR-03	Reliability	Database transactions maintain strict ACID compliance to prevent duplicate slot bookings.	99.9% data consistency
NFR-04	Usability	Responsive web interface designed for smooth interaction across desktop and mobile devices.	Clean, 3-tap workflow
NFR-05	Maintainability	Modular 3-tier architecture separating React frontend, Express API routes, and PostgreSQL storage.	Separation of concerns

4. REQUIREMENTS TRACEABILITY MATRIX

Req ID	Requirement Summary	User Role	Implementation Component
FR-01	Role-Based Authentication	All Roles	`/api/auth` (JWT + bcryptjs)
FR-02 / FR-09	Donor Profile & Eligibility	Donor	`users` table & eligibility calculation
FR-03 / FR-06	Urgent Requests & Privacy	Hospital / Requester	`blood_requests` table & APIs
FR-04 / FR-05	Matching Engine & Alerts	Donor & Requester	Matching log service & query engine
FR-07 / FR-08	Camp Creation & Booking	Organizer & Donor	`donation_camps` & `camp_bookings`

Academic Verification & Submission: This Software Requirements Specification has been compiled for academic review under Course CT-365 (Software Engineering) at Thar Institute of Engineering, Sciences and Technology (TIEST), NED University of Engineering & Technology.